Transfer: thermal energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose an energy-transfer model and communicate its limits.

SUCCESS CRITERIA - I can...

1. Distinguish warming, phase change and work.
2. Use correct signs and units.
3. Explain why a model result depends on its assumptions.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “phase change” and “internal energy” to explain your model or result. Sentence starters are optional.

phase change: a change between states such as solid and liquid

internal energy: energy stored in a system’s microscopic motion/interactions

efficiency: useful output divided by input

01 Bring back an earlier idea

A heat engine receives 500 J heat and outputs 150 J work per cycle.
Find efficiency and rejected heat.

Word help: cycle = a process that returns a system to its initial state

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $Q = mc\Delta T$; $Q = mL$; $\Delta U = Q - W_{by}$; $\eta = W_{out}/Q_{hot}$. Select the applicable model and supplied constants. State heat-loss, phase-change and cycle assumptions.

Worked reasoning: A system takes 120 J of heat, does 50 J of work and increases internal energy by 70 J. An energy model must include transfers and storage rather than treating all heat as useful work.

Your turn: Draw its system boundary, input, output and storage change.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

A hot block and cool water are isolated together.
Draw initial heat flow and describe equilibrium without inventing a final temperature.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Hot block



Cool water

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg sample with c=100 J/(kg·°C) warms by 4°C.
Find Q. A second sample changes phase with m=2 kg, L=300 J/kg; find its Q.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

An engine absorbs 500 J and produces 150 J work in a cycle.
Find rejected heat and explain why an energy balance alone does not establish that any temperature pair can achieve that efficiency.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A device transfers 24 J in 4 s, then performs the same transfer in 8 s.
Find both average powers and explain the difference.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

A system absorbs 40 J and does 15 J work. FindΔU using the unit convention.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Design a second process with the same 70 J stored increase but a different work output.
Explain why internal-energy change alone cannot determine heat input.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3